

SQL Data Warehouse and Shipment Analytics

Clean Portfolio Report

Executive Summary

This project builds a small Oracle SQL data warehouse for shipment analytics. The work demonstrates dimensional modelling, fact and dimension table design, Oracle SQL table creation, data population, key constraints and business-oriented analytical queries. The warehouse supports analysis of shipment volume, monthly shipment value, product performance, shipper performance and customer value.

Business Process and Grain

The selected business process is shipment performance analysis. The grain of the fact table is one shipment line: one product shipped to one customer on one date through one shipper. This level of detail allows the business to aggregate results by date, customer, product and shipping provider while preserving transactional detail for flexible analysis.

Dimensional Model

Table	Role	Main Fields
SHIP_FACT	Fact table	Shipment ID, date ID, customer ID, shipper ID, product ID, quantity, shipment weight, ship cost
DIM_CUSTOMER	Customer dimension	Customer ID, name, city, state
DIM_PRODUCT	Product dimension	Product ID, product name, product cost
DIM_SHIPPER	Shipper dimension	Shipper ID, shipper name, city, state
DIM_DATE	Date dimension	Date ID, shipment date

SQL Implementation

The Oracle SQL implementation creates the four dimension tables and the shipment fact table, then populates the fact table with generated shipment records for 2024. A separate enrichment script adds product names, primary keys, foreign keys and a sample business record. This keeps the project readable and separates setup logic from warehouse-integrity logic.

```
ALTER TABLE ship_fact ADD CONSTRAINT fk_ship_fact_customer
FOREIGN KEY (customer_id) REFERENCES dim_customer (customer_id);
```

Analytical Queries

The project includes SQL queries that answer practical business intelligence questions using joins, aggregation, date filtering and ranking. Examples include identifying the top shipper by shipment count, monthly shipment values, the best product in a given month, the strongest shipper by month and the highest-value customer.

```
SELECT s.ship_name, COUNT(f.shipment_id) AS total_shipments
FROM ship_fact f
JOIN dim_shipper s ON f.shipper_id = s.shipper_id
JOIN dim_date d ON f.date_id = d.date_id
```

```
WHERE EXTRACT(YEAR FROM d.ship_date) = 2024
GROUP BY s.ship_name
ORDER BY total_shipments DESC
FETCH FIRST 1 ROWS ONLY;
```

Key Findings from the Original Analysis

- The highest shipment count in the original analysis was produced by Cutesy Shipping, with 2,075 shipments in 2024.
- Monthly shipping cost analysis showed January as the strongest month in the original output, while April was the weakest month.
- A value-based calculation using quantity multiplied by product cost provided a more complete view of shipment value than shipping cost alone.
- The highest-value product in May 2024 was the product mapped to the lotion category in the original project data.
- The top customer analysis demonstrated how the star schema can support customer value reporting using fact-to-dimension joins.

Skills Demonstrated

- Dimensional modelling and star schema design
- Oracle SQL table creation and data population
- Primary key and foreign key constraint design
- Analytical SQL using joins, aggregation, date functions and ranking
- Business intelligence interpretation and reporting
- Portfolio-safe documentation and data-governance awareness

Portfolio Cleaning Note

This report is a cleaned portfolio version of a coursework project. Personal identifiers, assignment cover sheets, AI declarations, submission screenshots and private Oracle APEX workspace details have been removed. The SQL files in the repository have been converted into clean standalone scripts suitable for portfolio review.